

Design a Web Crawler - URL Frontier

1. What is URL Frontier?

- URL Frontier is a **data structure that stores URLs to be downloaded**
- It is a **core component** of the crawler

Responsibilities:

- Politeness
- URL prioritization
- Freshness

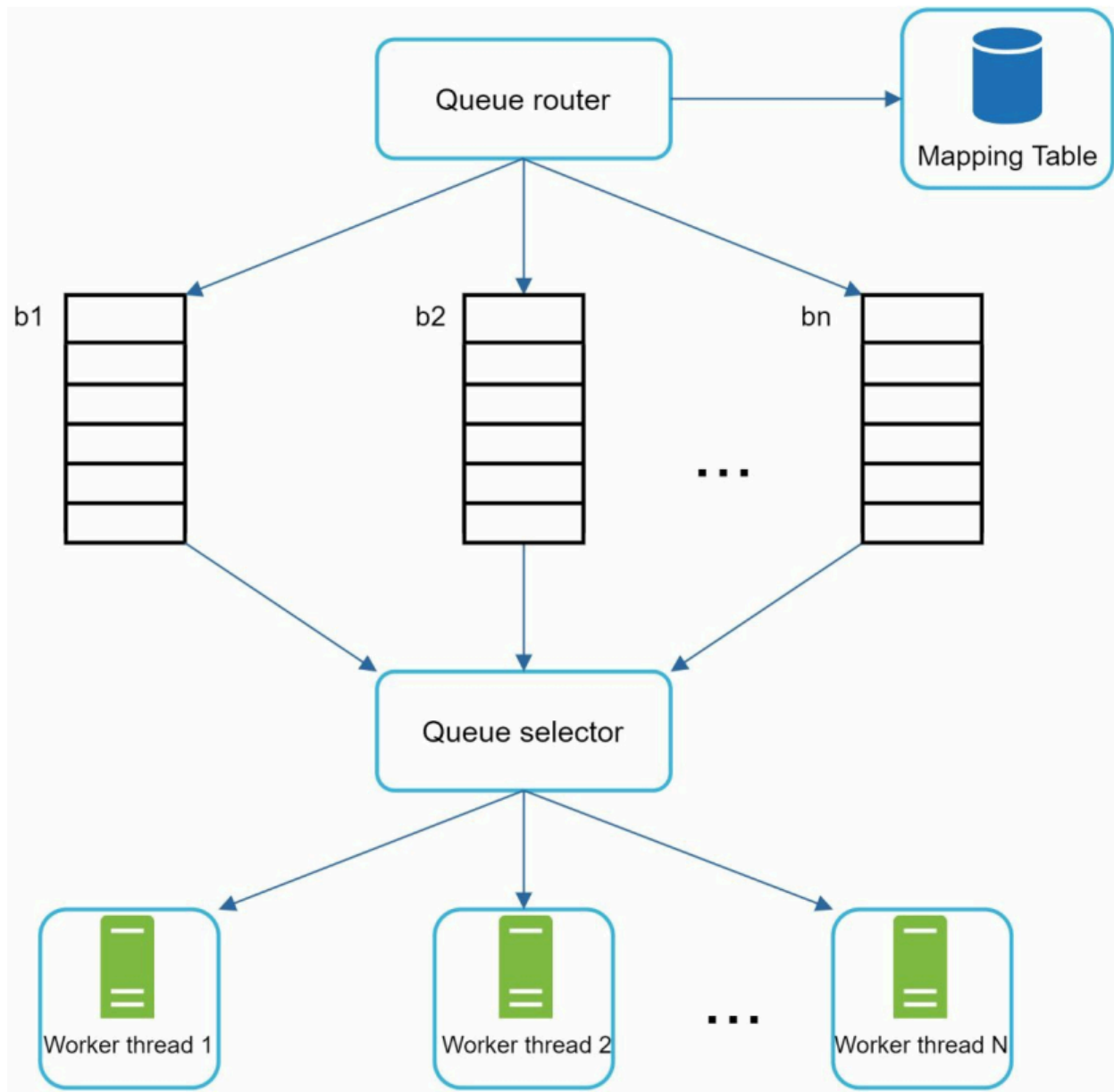
2. Politeness

Problem

- Sending too many requests to the same server:
 - Overloads servers
 - Considered as **DoS attack**

Solution

- Download **one page at a time per host**
- Add **delay between requests**



Components:

- **Queue Router**
 - Ensures each queue contains URLs from the **same host**
- **Mapping Table**
 - Maps host → queue

Host	Queue
wikipedia.com	b1
apple.com	b2
...	...
nike.com	bn

- **FIFO Queues (b1, b2, ... bn)**
 - Each queue stores URLs of a **single host**
- **Queue Selector**
 - Assigns queue to worker threads
- **Worker Threads (1 to N)**
 - Each thread:
 - Processes one host queue
 - Downloads sequentially
 - Applies delay

Flow

None

URL → Queue Router → Mapping Table → Host Queue → Worker Thread
→ Download (with delay)

3. URL Prioritization

Problem

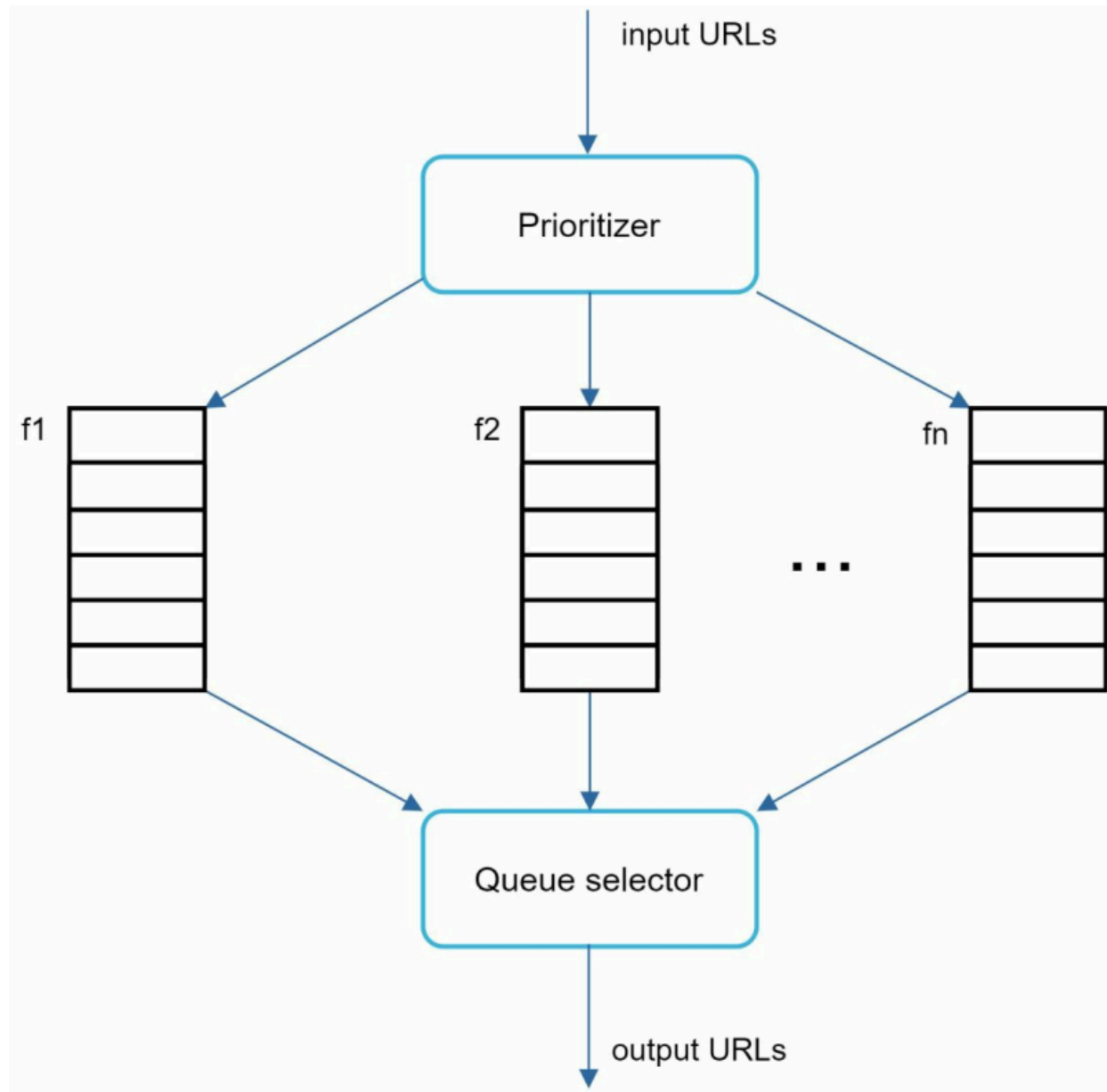
- Not all URLs are equally important

Solution

- Assign priority based on usefulness

Priority Factors

- PageRank
- Website traffic
- Update frequency



Components:

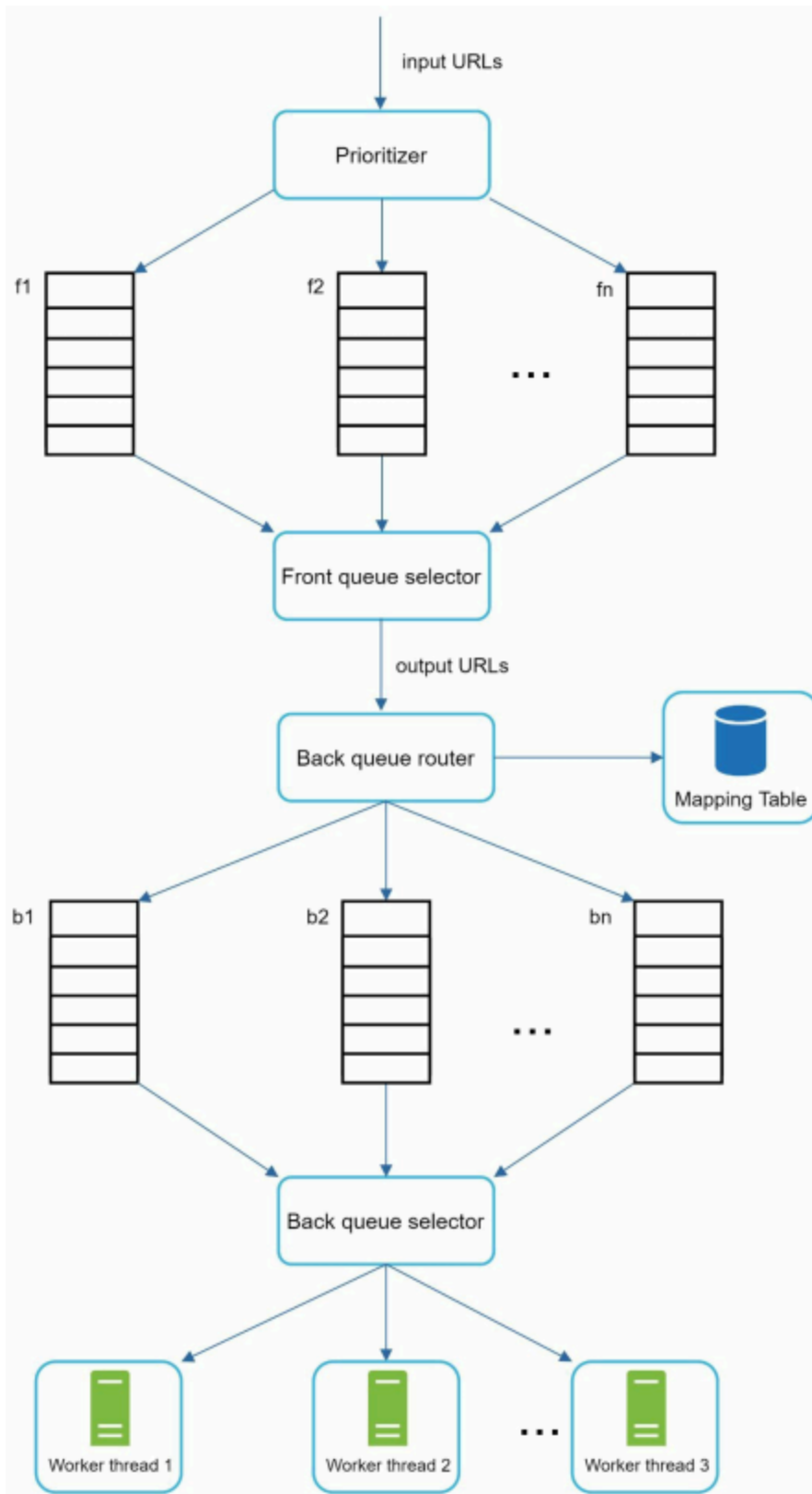
- **Prioritizer**
 - Assigns priority to URLs
- **Priority Queues (f1 to fn)**
 - Each queue represents a priority level
- **Queue Selector**
 - Picks queues with bias toward higher priority

Flow

None

URLs → Prioritizer → Priority Queues → Queue Selector → Worker
Threads

4. Combined URL Frontier Design



Two Main Modules:

Front Queues

- Manage **prioritization**

Back Queues

- Manage **politeness**

Flow

None

Incoming URLs



Front Queues (Priority)



Back Queues (Host-based)



Worker Threads



Download

5. Freshness

Problem

- Web pages are constantly:
 - Added
 - Updated
 - Deleted

Strategies

- Recrawl based on:
 - Update history
- Prioritize:
 - Important pages

6. Storage for URL Frontier

Problem

- Frontier size can be **hundreds of millions of URLs**

Memory Only

- Fast but not scalable

Disk Only

- Scalable but slow

Hybrid Approach

Storage:

- Disk → majority of URLs
- Memory → buffer for operations

Optimization:

- Periodic batch writes

Flow

None

Memory Buffer ⇌ Disk Storage

7. Key Takeaway

URL Frontier ensures:

- Controlled crawling (**politeness**)
- Efficient crawling (**priority**)
- Updated data (**freshness**)